Final Report

Telecom Customer Churn Analysis

1. Project Overview

Customer churn is a major issue for telecommunications companies. When customers cancel their service and move to a different provider, it directly affects the company's revenue and long-term growth. Because of this, telecom companies are increasingly using data analytics and machine learning techniques to understand customer behaviour and identify patterns that lead to churn.

The main goal of this project was to analyse telecom customer data and develop a model that can predict which customers are more likely to leave the service. By identifying these customers early, companies can take proactive steps to improve retention and customer satisfaction.

The project was completed in three main stages. First, the dataset was prepared by cleaning the data, handling missing values, and converting categorical variables into a suitable format for analysis. The second stage focused on customer segmentation using the K-Means clustering algorithm to group customers with similar characteristics. In the final stage, an Artificial Neural Network (ANN) model was developed to predict customer churn.

By combining clustering and predictive modelling, this project aims to provide meaningful insights that can help telecom companies better understand their customers and design effective retention strategies.

2. Customer Segmentation Findings

Customer segmentation was carried out using the **K-Means clustering algorithm**, which groups customers based on similarities in their data. Before applying the clustering model, the dataset was prepared by encoding categorical variables and scaling numerical values to ensure that all features contributed equally to the clustering process.

To determine the appropriate number of clusters, the **Elbow Method** was used. This technique helps identify the point where increasing the number of clusters no longer significantly improves the model.

After analysing the clustering results, several distinct groups of customers were identified. These clusters represent different types of telecom users. For example, some customers had long service tenures and stable usage patterns, suggesting they are loyal customers. Another group consisted of newer customers with shorter tenure, who may still be exploring different service providers.

The clustering analysis also revealed a segment of customers who appear more likely to churn. These customers typically have shorter contracts, higher monthly charges, or less commitment to the service.

Understanding these different customer segments allows telecom companies to better tailor their services and marketing strategies. For example, customers with shorter tenure may benefit from loyalty incentives, while high-value customers may require additional service benefits to maintain satisfaction.

3. Churn Prediction using Artificial Neural Networks (ANN)

In addition to customer segmentation, a predictive model was developed to estimate the likelihood of customer churn. For this purpose, an **Artificial Neural Network (ANN)** was implemented.

Artificial Neural Networks are commonly used in machine learning because they can detect complex relationships within data. In this project, the ANN model was trained using several important customer attributes, including tenure, monthly charges, contract type, internet service type, payment method, and demographic characteristics.

The architecture of the model included an **input layer**, which receives the customer features, followed by **two hidden layers** that process the data using ReLU activation functions. The final **output layer** uses a sigmoid activation function to produce a probability value indicating whether a customer is likely to churn.

The model was trained using the **Adam optimisation algorithm** and the **binary cross-entropy loss function**, which is suitable for binary classification problems.

After training the model on the dataset, the final model achieved an accuracy of approximately **78%**. While this is not perfect, it indicates that the model is capable of identifying general churn patterns within the data.

To evaluate the model's performance, several visualisations were generated, including an accuracy graph, a loss graph, and a confusion matrix. These metrics helped assess how well the model distinguishes between customers who remain with the company and those who leave.

4. Factors Contributing to Customer Churn

The analysis conducted during this project highlighted several factors that appear to influence customer churn.

One of the most significant factors is the **type of contract** a customer has. Customers with month-to-month contracts tend to have a higher churn rate compared to those who are committed to longer-term contracts. This suggests that contract flexibility may increase the likelihood of customers switching providers.

Another factor identified is **customer tenure**. Customers who have recently joined the service tend to churn more frequently than long-term customers. This may indicate that early customer experience plays an important role in retention.

Monthly service charges also appear to influence churn behaviour. Customers paying higher monthly fees may be more likely to explore alternative providers offering lower prices or better value.

Overall, these findings suggest that pricing strategies, contract structures, and early customer engagement are key elements in reducing churn.

5. Recommendations for Customer Retention Strategies

Based on the insights obtained from this analysis, several strategies can be recommended to help telecom companies improve customer retention.

One possible approach is to encourage customers to move from month-to-month contracts to longer-term plans by offering discounts or additional benefits. Long-term contracts provide more stability and reduce the likelihood of customers switching providers.

Another recommendation is to provide loyalty rewards or incentives for new customers during the early stages of their service. Improving the onboarding experience and offering early engagement benefits may help build stronger customer relationships.

Predictive models such as the ANN model developed in this project can also be used to identify customers who are at higher risk of churn. These customers can then be targeted with personalised offers, improved service packages, or customer support interventions.

By implementing these strategies, telecom companies can improve customer satisfaction while also reducing churn rates.

6. Limitations of the Study

While the project successfully demonstrates the use of machine learning techniques for churn prediction, there are several limitations that should be considered.

Firstly, the dataset used for this project is relatively limited and may not fully represent the diversity of real-world telecom customers. Larger datasets would likely improve the accuracy and reliability of the predictive model.

Secondly, there may be **class imbalance** in the dataset, where the number of non-churn customers significantly exceeds churn customers. This can influence model performance and prediction accuracy.

Finally, the ANN model architecture used in this project was relatively simple. With further experimentation and tuning, the model's performance could potentially be improved.

Future work could involve experimenting with different machine learning models such as **Random Forest, Gradient Boosting, or XGBoost**, as well as applying hyperparameter optimisation techniques.

7. Conclusion

This project demonstrates how machine learning techniques can be applied to analyse telecom customer data and predict customer churn.

Through the use of clustering analysis, the project identified several distinct customer segments with different service behaviours. These insights can help telecom companies better understand their customer base.

The Artificial Neural Network model developed in this project was able to predict customer churn with reasonable accuracy, providing a useful tool for identifying customers at risk of leaving the service.

Overall, the results highlight the potential of data-driven approaches in supporting customer retention strategies. By combining customer segmentation and predictive modelling, telecom companies can make more informed decisions and improve long-term customer relationships.